Growth as a Stochastic Process: From Observed *U* and *V* to Mean *SGPc*

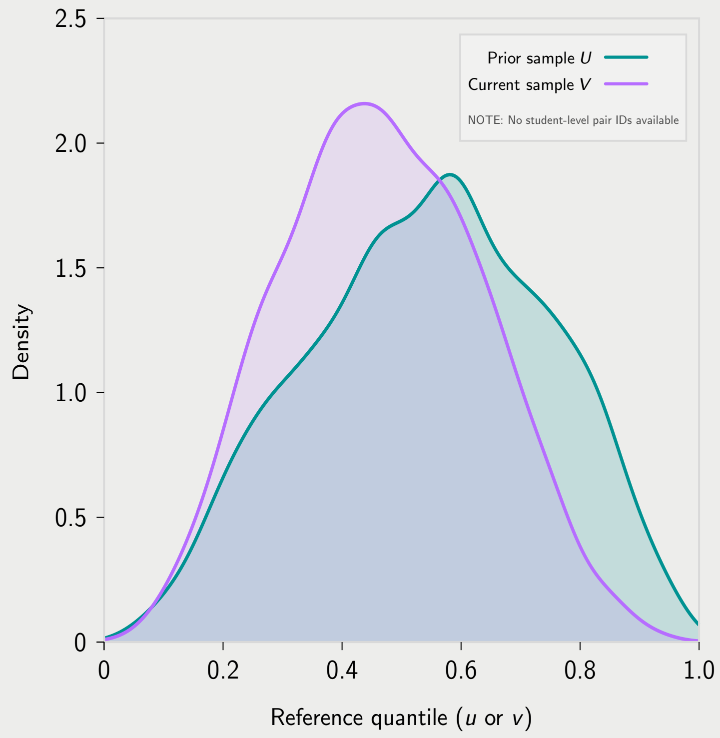
Synthetic low-growth case (true mean *SGPc* 39.0), baseline t-copula kernel ($\rho = 0.72$, df = 8)

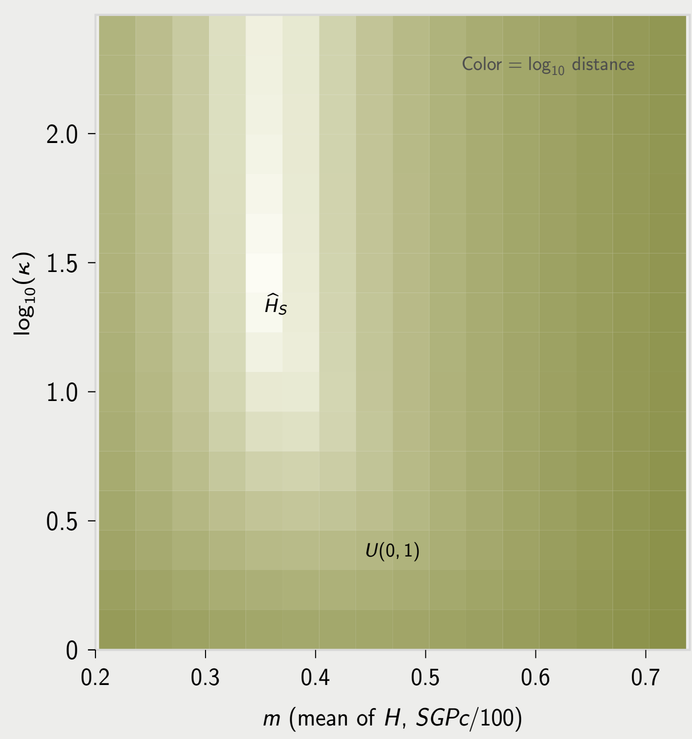
$W_1(F^{\text{obs}}_{V,S},\, F_H)$: Uniform = 0.0915, Best-fit = 0.0019 (97.9% reduction)

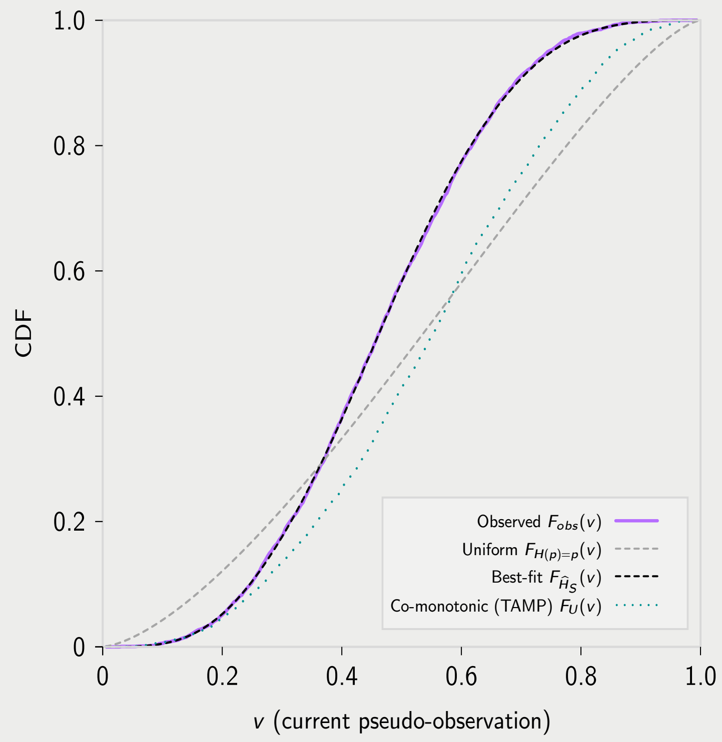
A. Map Scores to Pseudo-Observations
Observed marginals: U and V (no pairs)

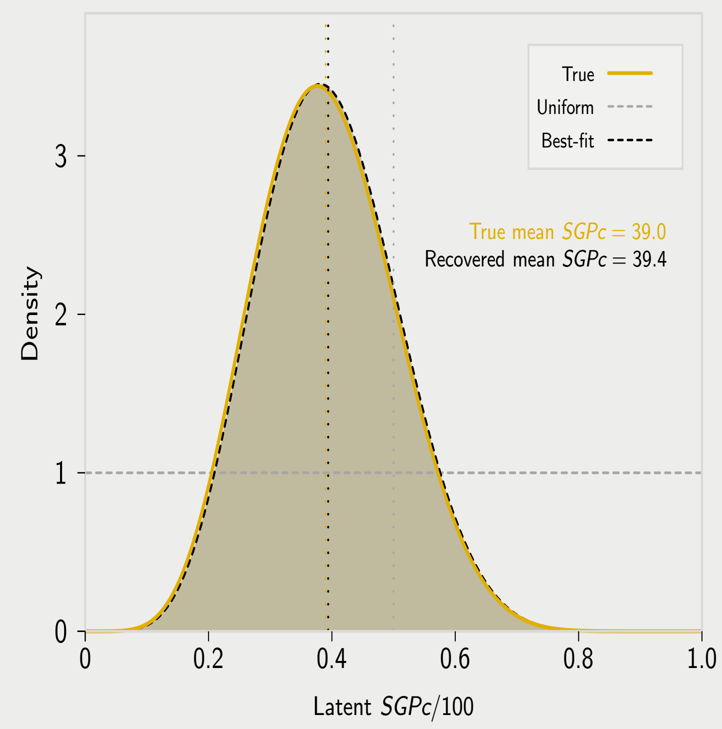
B1 & B2. Fit stochastic growth regime using canonical copula and observed *U* and *V*
Minimize Wasserstein-1 distance over $\text{Beta}(\kappa m, \kappa(1-m))$ between induced $F_H(v)$ and observed $F_{\text{obs}}(v)$

C. Recover growth regime P_S (latent SGPc)
SGPc distribution based upon fit CDF









We observe two independent samples: prior- and current-scores, X and Y, respectively, for the same subgroup S (e.g., a TIMSS country or a NAEP state). We map scores to reference-quantiles using fixed reference marginals:

1. Prior reference quantiles $U = F_X^{\text{ref}}(x_i)$ and
2. Current reference quantiles $V = F_Y^{\text{ref}}(y_j)$.

Crucially, there is *no student-level linkage* between these samples—we cannot match any u_i to its corresponding v_j.

Panel A shows the two *univariate* marginals plotted on the same reference-quantile axis (“u or v”). This is the situation faced by TIMSS/NAEP-style designs when cohorts are sampled independently.

The distributions U and V are the only observables. All subsequent inference proceeds from these marginals plus an externally supplied copula dependence structure.

We estimate the subgroup’s best-fit growth regime by searching over an admissible class $\mathscr{H}$ of candidate CDFs on $[0,1]$. In this panel, $\mathscr{H}$ is the *Beta regime family*

$$\mathscr{H}_{\text{Beta}} = \{H_{m,\kappa}(\cdot)\ :\ P \sim \text{Beta}(\kappa m, \kappa(1-m)),\ 0 < m < 1,\ \kappa > 0\},$$

where $m = \mathbb{E}[P]$ controls the regime center (mean SGPc on the 0–1 scale) and κ controls concentration (dispersion of latent SGPc).[2]

For each candidate $H \in \mathscr{H}$, Panel B2 provides the induced *predicted* current-grade marginal

$$F_H(v) \;=\; \mathbb{E}_U\left[H(F_0(v \mid U))\right].$$

The growth regime estimate for subgroup S is defined by the minimum-distance problem

$$\hat{H}_S \;=\; \underset{H \in \mathscr{H}_{\text{Beta}}}{\operatorname{argmin}}\ W_1(F^{\text{obs}}_{V,S},\, F_H),$$

where

$$W_1(F,G) = \int_0^1 \left|F(v) - G(v)\right| dv$$

is the Wasserstein-1 (earth mover’s) distance between the observed and predicted current-grade marginal CDFs.

The heat map displays $\log_{10}$ Wasserstein–1 distance over $(m, \log_{10}\kappa)$: lighter regions indicate candidate regimes whose induced F_H more closely matches $F^{\text{obs}}_{V,S}$. The marked optimum corresponds to $\hat{H}_S$ (equivalently, $(\hat{m}, \hat{\kappa})$). For reference, the uniform regime $H(p) = p$ occurs at $m = 0.5$ and $\kappa = 2$ (i.e., Beta(1, 1)), indicated by the $U(0,1)$ marker.

Let C_0 be a baseline copula (empirical or parametric) calibrated from longitudinal data. C_0 provides a *dependence template* for how students move from U to V and induces the conditional CDF (transition kernel)

$$F_0(v \mid u) \;=\; \Pr(V \le v \mid U = u) \;=\; \frac{\partial}{\partial u} C_0(u, v).$$

Its inverse in v, the conditional quantile function/kernel, is

$$Q_0(p \mid u) = F_0^{-1}(p \mid u).$$

Intuitively, for each starting quantile u, the kernel $F_0(\cdot \mid u)$ describes the distribution of possible ending quantiles V under the baseline dependence structure C_0.

Let $P \in [0,1]$ be a random variable (i.e., a latent SGPc on the 0–1 scale) and p denote a realization in $[0,1]$. A *growth regime*, H, is the distribution of P represented by its CDF:

$$H(p) \;:=\; \Pr(P \le p), \qquad p \in [0,1].$$

Given $U = u$ and a draw $P = p$, the model generates an outcome percentile by[1]

$$V \;=\; Q_0(P \mid U = u).$$

A growth regime governs how the subgroup S “occupies” the baseline kernel $F_0(\cdot \mid u)$. The baseline growth regime is Uniform(0, 1) and its CDF is the identity on $[0,1]$. That is,

$$P \sim \text{Uniform}(0,1) \quad\Longleftrightarrow\quad H(p) = p.$$

Departures from $H(p) = p$ tilt mass toward lower or higher conditional percentiles relative to the uniform baseline regime. Although we do not observe pairs (u_i, v_i) in TIMSS- or NAEP-style settings, the model provides a *forward map* from the prior to current margin.

Thus, for any fixed u,

$$\begin{aligned}
\Pr(V \le v \mid U = u) &= \Pr\big(Q_0(P \mid u) \le v\big) \\
&= \Pr\big(P \le F_0(v \mid u)\big) \\
&= H\big(F_0(v \mid u)\big).
\end{aligned}$$

Therefore, by the law of total probability, the subgroup’s predicted current-grade marginal CDF is

$$\begin{aligned}
F_H(v) &= \Pr(V \le v) \\
&= \mathbb{E}_U\left[H(F_0(v \mid U))\right] \\
&= \int_0^1 H\big(F_0(v \mid u)\big)\, dF_U(u) \\
&\approx \frac{\sum_{i=1}^n w_i\, H\big(F_0(v \mid u_i)\big)}{\sum_{i=1}^n w_i},
\end{aligned}$$

where $\{u_i, w_i\}$ are the observed prior pseudo-observations and survey weights for the subgroup (i.e., a weighted empirical approximation to F_U).

The output of Panel B1 is a subgroup-specific *growth regime* H_S, where H_S is a cumulative distribution function (CDF) on $[0,1]$ for a latent conditional-percentile random variable P_S (SGPc on the 0–1 scale):

$$H_S(p) \;=\; \Pr(P_S \le p), \qquad 0 \le p \le 1.$$

In words, H_S describes how subgroup S allocates conditional ranks *within* the baseline dependence template $F_0(\cdot \mid u)$.

Panel C shows the *distribution of* P_S (typically via its density $f_S(p) = H_S'(p)$ when H_S is smooth), not the CDF itself. Thus the “Best-fit” curve is the density corresponding to the fitted CDF H_S (i.e., $P_S \sim H_S$). The dashed grey “Uniform” reference corresponds to the uniform regime $H(p) = p$ (equivalently $P \sim \text{Unif}(0,1)$), which places no tilt on conditional percentiles relative to the baseline kernel.

Because we restrict H to the Beta family in Panel B1, $P_S \sim \text{Beta}(\kappa m, \kappa(1-m))$: the mean m sets the regime center (reported on the SGPc scale as $100m$) and κ controls concentration (dispersion of latent SGPc).

The vertical reference lines mark the mean SGPc under the uniform regime (50), the generating “True” regime (), and the recovered best-fit regime (). In applications, the primary outputs are H_S (and uncertainty bands) and summary functionals such as mean/median SGPc and bucket membership probabilities.

[1] We assume $P \perp U$. That is, p is drawn independently of starting point within the subgroup. This assumption is relaxed in the full model.
[2] Other regime families considered include the truncated exponential family (the maximum entropy probability distribution for a random variable restricted to a finite interval with a fixed mean) and the truncated uniform family.